WORKED SOLUTION

STEP III 2002, Question 1

Pure • Improper Integrals and Volumes

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Read the question before you go on. It is in the STEP Support Programme's question database:

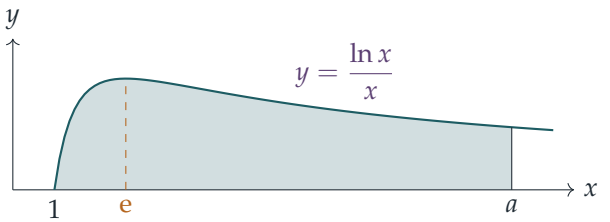
step.maths.org/questions/02-s3-q1

The key idea

Both integrals become easy once you notice that $\frac{1}{x}$ is the derivative of $\ln x$. The area integrand is "a function times its own derivative", so it integrates at sight. The volume integrand $(\ln x)^2/x^2$ has a leftover power of x, so we integrate by parts twice, each time differentiating a power of $\ln x$ to lower it. The interesting part is the contrast between the two limits: the curve decays just slowly enough for the area to grow without bound, but squaring it makes it decay fast enough for the volume to settle down.

Solution

For $x \geq 1$ we have $\ln x \geq 0$, so the curve lies on or above the x-axis and the area is simply the integral. It starts at $(1,0)$, rises to a maximum at $x = \mathrm{e}$ (where $y' = (1-\ln x)/x^2 = 0$), and then decays slowly towards the axis.



The area

Since $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}(\ln x)^2\right) = \dfrac{\ln x}{x}$,

$$A(a) = \int_1^a \frac{\ln x}{x}\,\mathrm{d}x = \left[\tfrac{1}{2}(\ln x)^2\right]_1^a = \tfrac{1}{2}(\ln a)^2,$$

because $\ln 1 = 0$. (Equivalently, substitute $u = \ln x$, $\mathrm{d}u = \mathrm{d}x/x$, to get $\int_0^{\ln a} u\,\mathrm{d}u$.)

As $a \to \infty$, $\ln a \to \infty$, so $A(a) \to \infty$: the area is unbounded, although it grows very slowly.

The volume

Rotating about the x-axis, the volume is $V(a) = \pi \int_1^a y^2 \,\mathrm{d}x = \pi \int_1^a \dfrac{(\ln x)^2}{x^2} \,\mathrm{d}x$.

Integrate by parts with $u = (\ln x)^2$ and $\mathrm{d}v = x^{-2}\,\mathrm{d}x$, so $\mathrm{d}u = \frac{2\ln x}{x}\,\mathrm{d}x$ and $v = -\frac{1}{x}$:

$$\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = -\frac{(\ln x)^2}{x} + 2\int \frac{\ln x}{x^2}\,\mathrm{d}x.$$

Doing the same again with $u = \ln x$, $\mathrm{d}v = x^{-2}\,\mathrm{d}x$:

$$\int \frac{\ln x}{x^2}\,\mathrm{d}x = -\frac{\ln x}{x} + \int \frac{1}{x^2}\,\mathrm{d}x = -\frac{\ln x}{x} - \frac{1}{x}.$$

Putting these together,

$$\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = -\frac{(\ln x)^2 + 2\ln x + 2}{x} + C.$$

At $x = 1$ the bracket is $0 + 0 + 2$, so the lower limit contributes $-(-2) = +2$, and

$$V(a) = \pi\left(2 - \frac{(\ln a)^2 + 2\ln a + 2}{a}\right).$$

> **Ask yourself** How do we know $(\ln a)^2/a$ tends to 0, when its top and bottom both tend to infinity?

Logarithms grow more slowly than any positive power. To make this precise, start from $\ln t < t$ for all $t > 0$ (the line $y = t$ lies above $y = \ln t$; the gap $t - \ln t$ has its minimum value 1 at $t = 1$). Put $t = a^{1/4}$: then $\frac{1}{4}\ln a < a^{1/4}$, that is $\ln a < 4a^{1/4}$. So for $a > 1$,

$$0 < \frac{(\ln a)^2}{a} < \frac{16\,a^{1/2}}{a} = \frac{16}{\sqrt{a}} \to 0, \qquad 0 < \frac{\ln a}{a} < \frac{4}{a^{3/4}} \to 0,$$

and of course $2/a \to 0$. Hence the fraction in $V(a)$ tends to 0, and

$$V(a) \to 2\pi \quad \text{as } a \to \infty.$$

Since the integrand is positive, $V(a)$ increases with a, so the volume rises steadily towards 2π but never reaches it: the solid has finite volume even though the region being rotated has infinite area.

Checks

- At $a = 1$ both formulae give 0, as they must: $A(1) = 0$ and $V(1) = \pi(2 - 2) = 0$.

- Differentiating $-\left((\ln x)^2 + 2\ln x + 2\right)/x$ gives back $(\ln x)^2/x^2$ (the $2\ln x$ and 2 terms cancel against the derivative of the bracket).
- Numerically, Simpson's rule on $\pi\int_1^{10}(\ln x)^2/x^2\,\mathrm{d}x$ gives 2.54247, matching the formula; and with $x = \mathrm{e}^t$ the infinite volume is $\pi\int_0^{\infty} t^2\mathrm{e}^{-t}\,\mathrm{d}t$, which evaluates numerically to $6.283185\ldots = 2\pi$.

Common slips

- Forgetting the lower limit: $\ln 1 = 0$ kills most terms at $x = 1$, but not the constant $-2/x$, which supplies the whole limiting value 2π.
- Asserting $(\ln a)^2/a \to 0$ without any reason. A one-line comparison such as $\ln a < 4a^{1/4}$ is enough.

Beyond STEP

This is a cousin of "Gabriel's horn" (rotating $y = 1/x$), a solid with finite volume but infinite surface area: here even the plane region has infinite area. The substitution $x = \mathrm{e}^t$ turns the volume into $\pi\int_0^{\infty} t^2\mathrm{e}^{-t}\,\mathrm{d}t = \pi\,\Gamma(3) = 2!\,\pi$, a first glimpse of the gamma function, $\Gamma(n+1) = n!$. Making sense of improper integrals, and of comparisons such as "logs lose to powers", is a core topic of first-year analysis; Abbott's *Understanding Analysis* is a good place to read about it.

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